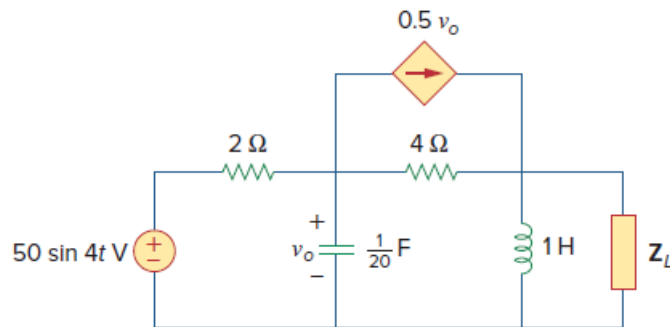


Problem

For the circuit in Fig. 11.47, find the value of Z_L that will receive the maximum power from the circuit. Then calculate the power delivered to the load Z_L .

Figure 11.47:



Step-by-step solution

Step 1 of 13

Refer to Figure 11.47 in the text book.

Consider the value of the voltage source v_s .

$$v_s = 50 \sin 4t \text{ V}$$

Compare with $v_m \sin \omega t$.

Therefore, the value of v_m and ω are 50 V and 4 rad/sec respectively.

Convert the value of the voltage to phasor form.

$$V_s = 50 \angle 0^\circ \text{ V}$$

Therefore, the value of the source voltage in phasor form is, $50 \angle 0^\circ \text{ V}$.

Step 2 of 13

Calculate the value of the reactance due to inductance.

$$X_L = j\omega L$$

Substitute 4 rad/sec for ω and 1 H for L .

$$\begin{aligned} X_L &= j(4)(1) \\ &= j4 \Omega \end{aligned}$$

Therefore, the value of the reactance X_L due to inductance is, $j4 \Omega$.

Step 3 of 13

Calculate the value of the reactance due to capacitance.

$$X_c = \frac{1}{j\omega C}$$

Substitute 4 rad/sec for ω and $\frac{1}{20}$ F for C .

$$\begin{aligned} X_c &= \frac{1}{j(4)\left(\frac{1}{20}\right)} \\ &= -j5 \Omega \end{aligned}$$

Therefore, the value of the reactance X_c due to capacitance is, $-j5 \Omega$.

Step 4 of 13

Consider the circuit diagram shown in Figure 1 to calculate the open circuit voltage.

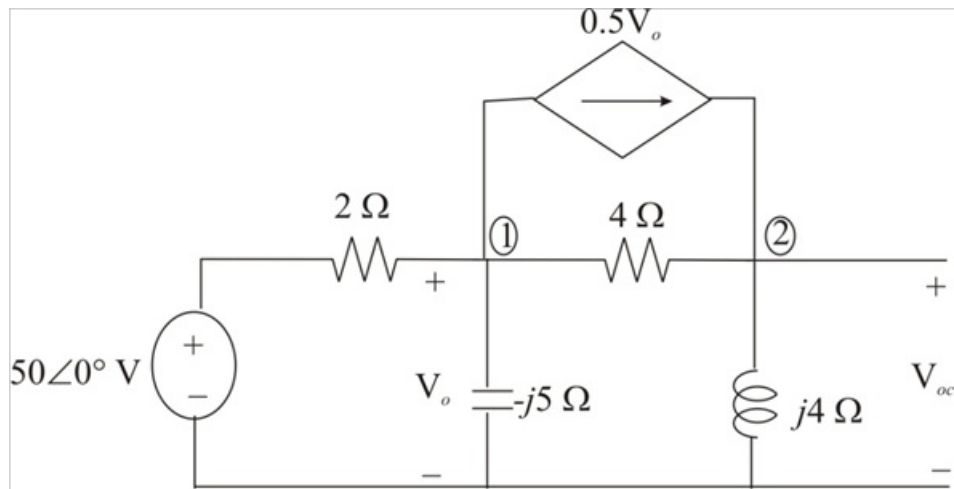


Figure 1

Step 5 of 13

Apply Kirchhoff's current law at node 1.

$$\frac{V_o - 50\angle 0^\circ}{2} + \frac{V_o}{-j5} + \frac{V_o - V_{oc}}{4} + 0.5V_o = 0$$

$$V_o \left(\frac{1}{2} - \frac{1}{j5} + \frac{1}{4} + 0.5 \right) - \frac{50\angle 0^\circ}{2} - \frac{V_{oc}}{4} = 0$$

$$V_o (0.5 + j0.2 + 0.25 + 0.5) - 25\angle 0^\circ - 0.25V_{oc} = 0$$

$$V_o (1.25 + j0.2) = 25\angle 0^\circ + 0.25V_{oc}$$

$$V_o = \frac{25\angle 0^\circ + 0.25V_{oc}}{1.266\angle 9.09^\circ}$$

$$= 19.747\angle -9.09^\circ + (0.198\angle -9.09^\circ)V_{oc}$$

Therefore, the expression for the voltage V_o is, $19.747\angle -9.09^\circ + (0.198\angle -9.09^\circ)V_{oc}$.

Step 6 of 13

Apply Kirchhoff's current law at node 2.

$$\frac{V_{oc}}{j4} + \frac{V_{oc} - V_o}{4} - 0.5V_o = 0$$

$$V_{oc} \left(\frac{1}{j4} + \frac{1}{4} \right) - V_o \left(\frac{1}{4} + 0.5 \right) = 0$$

$$V_{oc} (0.25 - j0.25) - V_o (0.25 + 0.5) = 0$$

$$V_{oc} (0.25 - j0.25) - V_o (0.75) = 0$$

$$V_{oc} (0.354 \angle -45^\circ) = V_o (0.75)$$

$$V_{oc} = V_o \left(\frac{0.75}{0.354 \angle -45^\circ} \right)$$

$$V_{oc} = V_o (2.119 \angle 45^\circ) \dots\dots (1)$$

Step 7 of 13

Recall equation (1).

$$V_{oc} = V_o (2.119 \angle 45^\circ)$$

Substitute $19.747 \angle -9.09^\circ + (0.198 \angle -9.09^\circ) V_{oc}$ for V_o .

$$V_{oc} = (19.747 \angle -9.09^\circ + (0.198 \angle -9.09^\circ) V_{oc}) (2.119 \angle 45^\circ)$$

$$V_{oc} = 41.844 \angle 35.91^\circ + (0.42 \angle 35.91^\circ) V_{oc}$$

$$V_{oc} - (0.42 \angle 35.91^\circ) V_{oc} = 41.844 \angle 35.91^\circ$$

$$(1 - 0.42 \angle 35.91^\circ) V_{oc} = 41.844 \angle 35.91^\circ$$

$$(0.704 \angle -20.47^\circ) V_{oc} = 41.844 \angle 35.91^\circ$$

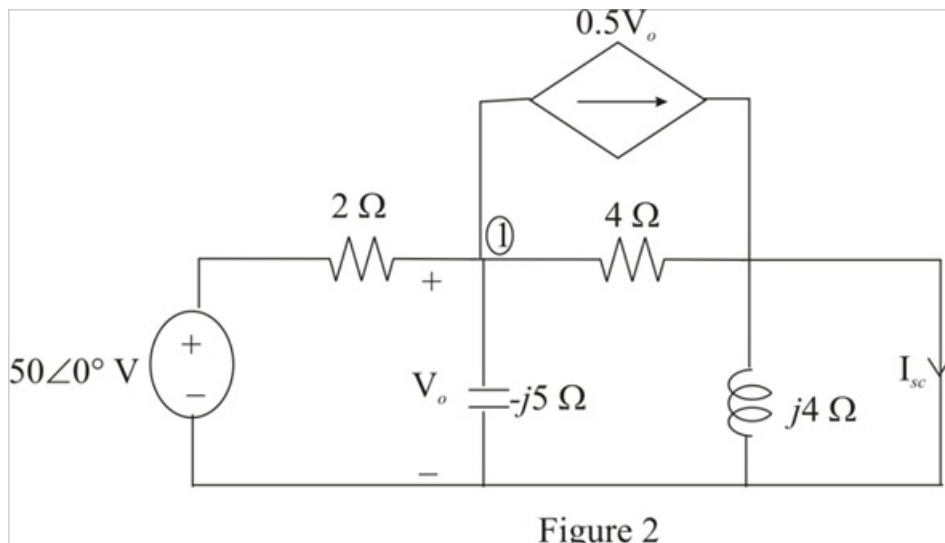
$$V_{oc} = \frac{41.844 \angle 35.91^\circ}{0.704 \angle -20.47^\circ}$$

$$V_{oc} = 59.438 \angle 56.38^\circ \text{ V}$$

Therefore, the value of the open circuit voltage V_{oc} is $59.438 \angle 56.38^\circ \text{ V}$.

Step 8 of 13

Short the output terminals as shown in Figure 2 to calculate the short circuit current.



Step 9 of 13

Notice that the inductor is shorted.

Apply Kirchhoff's current law at node 1.

$$\frac{V_o - 50\angle 0^\circ}{2} + \frac{V_o}{-j5} + \frac{V_o}{4} + 0.5V_o = 0$$

$$V_o \left(\frac{1}{2} - \frac{1}{j5} + \frac{1}{4} + 0.5 \right) - \frac{50\angle 0^\circ}{2} = 0$$

$$V_o (0.5 + j0.2 + 0.25 + 0.5) - 25\angle 0^\circ = 0$$

$$V_o (1.25 + j0.2) = 25\angle 0^\circ$$

$$V_o = \frac{25\angle 0^\circ}{1.266\angle 9.09^\circ} \\ = 19.747\angle -9.09^\circ \text{ V}$$

Therefore, the value of the voltage V_o is, $19.747\angle -9.09^\circ \text{ V}$.

Step 10 of 13

Calculate the value of the short circuit current I_{sc} .

$$I_{sc} = \frac{V_o}{4} + 0.5V_o \\ = 0.25V_o + 0.5V_o \\ = 0.75V_o$$

Substitute $19.747\angle -9.09^\circ \text{ V}$ for V_o .

$$I_{sc} = 0.75(19.747\angle -9.09^\circ) \\ = 14.81\angle -9.09^\circ \text{ A}$$

Therefore, the value of the short circuit current I_{sc} is, $14.81\angle -9.09^\circ \text{ A}$.

Step 11 of 13

Calculate the value of the Thevenin's impedance Z_{Th} .

$$Z_{Th} = \frac{V_{oc}}{I_{sc}}$$

Substitute $14.81\angle -9.09^\circ \text{ A}$ for I_{sc} and $59.438\angle 56.38^\circ \text{ V}$ for V_{oc} .

$$Z_{Th} = \frac{59.438\angle 56.38^\circ}{14.81\angle -9.09^\circ} \\ = 4.01\angle 65.47^\circ \Omega$$

Therefore, the value of the Thevenin's impedance Z_{Th} is, $4.01\angle 65.47^\circ \Omega$.

Step 12 of 13

Calculate the value of the load impedance that will receive the maximum power.

$$Z_L = (Z_{Th})^*$$

Substitute $4.01\angle 65.47^\circ \Omega$ for Z_{Th} .

$$Z_L = (4.01\angle 65.47^\circ)^* \\ = 4.01\angle -65.47^\circ \\ = 1.665 - j3.648 \Omega$$

Therefore, the value of the load impedance that will receive the maximum power is

$$\boxed{1.665 - j3.648 \Omega}.$$

Calculate the value of the maximum power absorbed by the load.

$$P_{\max} = \frac{|V_{oc}|^2}{8R_{Th}}$$

Substitute $1.665 \, \Omega$ for R_{Th} and $59.438 \, V$ for $|V_{oc}|$.

$$\begin{aligned} P_{\max} &= \frac{(59.438)^2}{8(1.665)} \\ &= \frac{3532.876}{13.32} \\ &= 265.23 \, W \end{aligned}$$

Therefore, the value of the maximum power $P_{\max}$ absorbed by load is, $265.23 \, W$.